



Name of the Student: _____

Sec :

Sub: MATHS

Time: 3 hours

Marks : 100

SUMMATIVE ASSESSMENT 2**Section - I****I. Answer the questions in one word or phrase.****12 x 1 = 12****Each question carries 1 mark.**1. Simplest rationalising factor of $\sqrt{8}$ is _____

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a. $\sqrt{8}$

b. $\sqrt{2}$

c. $2\sqrt{2}$

d. $2\sqrt{3}$

2. The zero of the polynomial $2x + 1$ is _____

[]

a. $\frac{1}{2}$

b. 2

c. $-\frac{1}{2}$

d. -2

3. Angles in the same segment of a circle are _____

4. The position of (2,3) and (3,20) represent same on the graph paper. True / False

5. How many books are there in Euclid's ancient Greek text on geometry, Elements?

a. 10

b. 20

c. 15

d. 13

6. The figure formed by joining the midpoints of sides of a quadrilateral successively is _____

[]

a. Parallelogram

b. Rhombus

c. Square

d. rectangel

7. Assertion (A) : The complementary angle of 60° is 30° .Reason (R) : The complementary angle of x° is $90^\circ - x^\circ$

A. Both Assertion and Reason are correct and Reason is the correct explanation for Assertion.

B. Both Assertion and Reason are correct and Reason is not the correct explanation for Assertion.

C. Assertion is true but the Reason is false.

D. Assertion is false but the Reason is true.

8. Match the following.

I. Curved surface area of cone

[]

a. $\pi r l$

II. Volume of sphere

[]

b. $3\pi r^2$

III. Total surface area of Hemisphere

[]

c. $\frac{2}{3}\pi r^3$

d. $\frac{4}{3}\pi r^3$

a. I-c, II-d, III-a

b. I-a, II-d, III-b

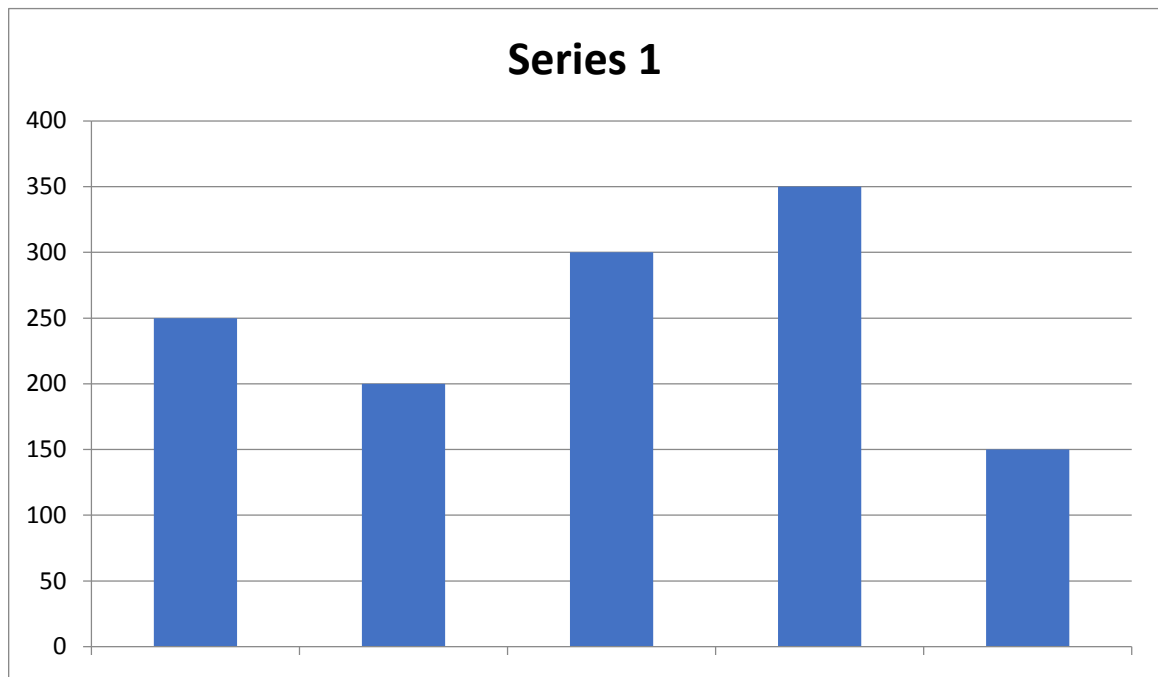
c. I-d, II-b, III-c

d. I-d, II-b, III-a

9. $3x + 7y - 3 = 0$ is a

- a. Simple equation b. Linear polynomial c. Quadratic polynomial d. Linear equation

10. The name of the graph is _____



11. In the Heron's formula which of the following is correct? []

- a. S is the perimeter of triangle. c. $2S$ is semi-perimeter of triangle.
b. S is semi-perimeter of triangle. d. $\frac{S}{2}$ is perimeter of triangle.

12. The angle at the centre of the circle is _____ []

- a. 0° b. 90° c. 360° d. 180°

Section - II

II. Answer all questions

Each question carries 2 marks.

8 x 2 = 16

13. Evaluate 104×96 by using suitable identity.

14. Find the each angle of equilateral triangle.

15. Find the volume of a sphere whose base diameter is 14cm.

16. Examine whether $x + 2$ is a factor of $x^3 + 3x^2 + 3x + 1$.

17. Write any two Euclid's postulates.

18. Write the heron's formula and explain the each term in it.

19. Find the value of K , if $x = 2$, $y = 1$ is a solution of the equation $2x + 3y = K$.

20. Draw the neat diagram of the following.

- a. Right circular cone b. Cylinder

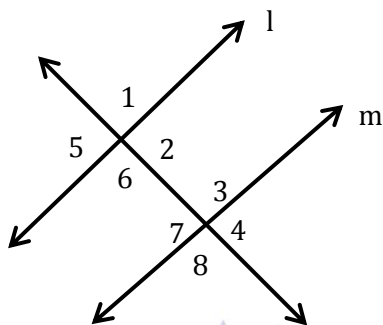
III. Answer all questions

8 x 4 = 32

Each question carries 4 marks.

21. Represent $\sqrt{2}$ on the number line.

22. Name the pair of angles represented by adjacent figure.



23. Write any four Euclid axioms.

24. Give one example of each of the following.

a. Binomial of degree 2

c. Linear polynomial in x and y

b. Monomial of degree 3

d. Quadratic polynomial in one variable 'y'

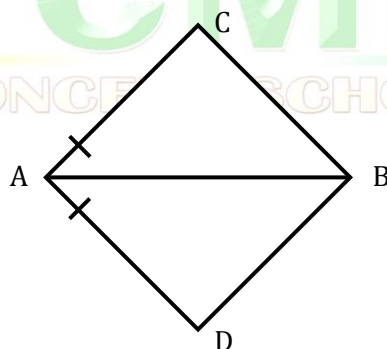
25. Write each of the following equations in the form $ax + by + c = 0$ and indicate the values of a, b and c in each case.

a. $2x + 3y = 4.37$

c. $2x = y$

b. $x - 4 = \sqrt{3}y$

d. $4 = 5x - 3y$

26. In quadrilateral ABCD, $AC = AD$ and AB bisects $\angle A$. Show that $\triangle ABC \cong \triangle ABD$ 

27. Prove that a cyclic parallelogram is a rectangle.

28. Curved surface area of a cone is 308 cm^2 and its slant height is 14cm. find

a. Radius of the base and

b. Total surface area of the cone.

Section - IV

IV. Answer all questions

5 x 8 = 40

Each question carries 4 marks.

29.A. Simplify the following expressions.

a. $(5 + \sqrt{7})(2 + \sqrt{5})$

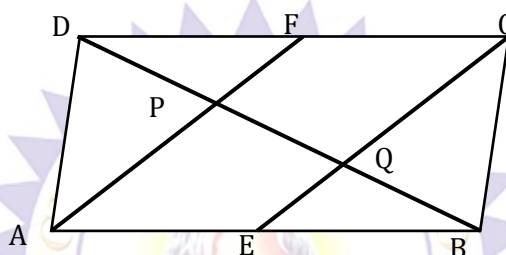
c. $(\sqrt{3} + \sqrt{7})^2$

b. $(5 + \sqrt{5})(5 - \sqrt{5})$

d. $(\sqrt{11} - \sqrt{7})(\sqrt{11} + \sqrt{7})$

(Or)

B. In a parallelogram $AB=CD$, E and F are the midpoints of sides AB and CD respectively. Show that the line segments AF and EC trisect the diagonal BD

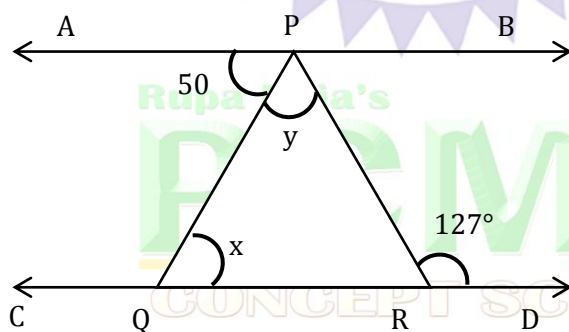


30.A. Verify that

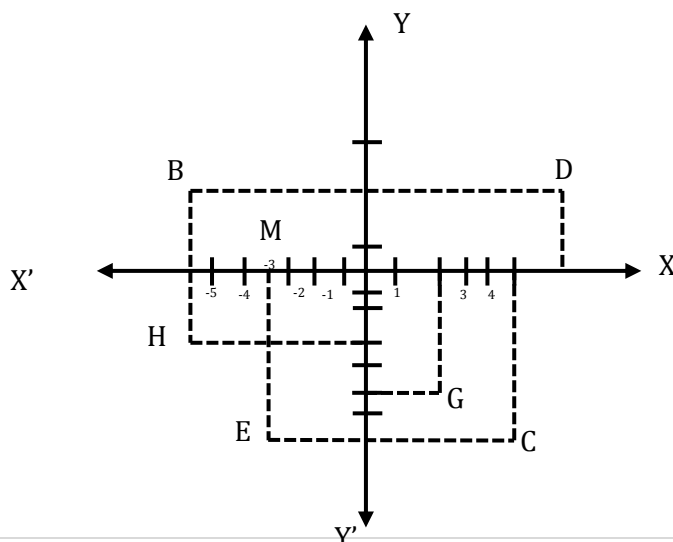
$$x^3 + y^3 + z^3 - 3xyz = \frac{1}{2}(x + y + z)[(x - y)^2 + (y - z)^2 + (z - x)^2]$$

(Or)

B. In the figure if $AB \parallel CD$, $\angle APQ = 50^\circ$ and $\angle PRD = 127^\circ$, find x and y .



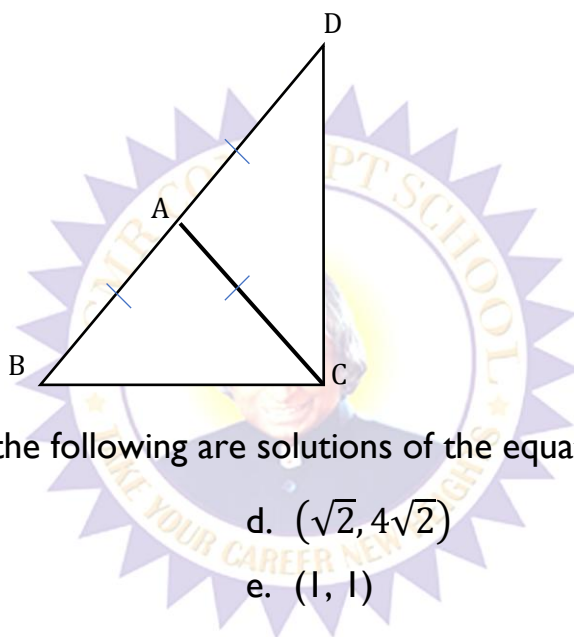
31.ZA. from the adjacent figure, Answer the following questions.



- i. The co-ordinates of B
- ii. The co-ordinates of C
- iii. The point identified by the co-ordinates (-3,-5)
- iv. The point identified by the co-ordinates (2, -4)
- v. The abscissa of the point D
- vi. The ordinate of the point H
- vii. The coordinates of the point L
- viii. The coordinates of the point M

(Or)

B. $\triangle ABC$ is an isosceles triangle in which $AB=AC$, side BA is produced to D such that $AD=AB$. Show that $\angle BCD$ is a right angle.



32.A. Check which of the following are solutions of the equations $x - 2y = 4$.

- | | | |
|-----------|----------------------------|------|
| a. (0, 2) | d. $(\sqrt{2}, 4\sqrt{2})$ | (Or) |
| b. (2, 0) | e. (1, 1) | |
| c. (4, 0) | | |

B. ABCD is a rhombus and P, Q, R and S are the midpoints of the sides AB, BC, CD and DA respectively, Show that the quadrilateral PQRS is a rectangle.

33.A. Represent the marks of the students of class IX on the graph by frequency polygon.

Marks	0-10	10-20	20-30	30-40	40-50
Frequency	3	9	17	12	9

(Or)

B. Prove that the angle subtended by an arc at the centre is double the angle subtended by it at any point on the remaining part of the circle.